1. Write a Java program that works as a simple calculator. Use a grid layout to arrange buttons for the digits and for the +, -,*, % operations. Add a text field to display the result. Handle any possible exceptions like divide by zero.

Program:-

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Calculator extends JFrame
{
   public Calculator()
   {
      setTitle("Calculator");
      setSize(300, 300);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      CalculatorPanel calc = new CalculatorPanel();
      add(calc);

      setVisible(true);
   }

   public static void main(String[] args)
   {
      new Calculator();
   }
}

class CalculatorPanel extends JPanel implements ActionListener
{
   JButton n1,n2,n3,n4,n5,n6,n7,n8,n9,n0,plus,minus,mul,div,dot,equal;

   static JTextField result = new JTextField("0",45);

   static String lastCommand = null;

   double preRes = 0, secVal = 0, res;

   private static void assign(String no)
   {
      if(result.getText().equals("0"))
         result.setText(no);
      else if(lastCommand == "=")
      {
         result.setText(no);
         lastCommand = null;
      }
      else
```

```
        result.setText(result.getText() + no);
    }

    public CalculatorPanel()
    {
        setLayout(new BorderLayout());

        result.setEditable(false);
        add(result, BorderLayout.NORTH);

        JPanel panel = new JPanel();
        panel.setLayout(new GridLayout(4,4));

        n7=new JButton("7"); panel.add(n7); n7.addActionListener(this);
        n8=new JButton("8"); panel.add(n8); n8.addActionListener(this);
        n9=new JButton("9"); panel.add(n9); n9.addActionListener(this);
        div=new JButton("/"); panel.add(div); div.addActionListener(this);

        n4=new JButton("4"); panel.add(n4); n4.addActionListener(this);
        n5=new JButton("5"); panel.add(n5); n5.addActionListener(this);
        n6=new JButton("6"); panel.add(n6); n6.addActionListener(this);
        mul=new JButton("*"); panel.add(mul); mul.addActionListener(this);

        n1=new JButton("1"); panel.add(n1); n1.addActionListener(this);
        n2=new JButton("2"); panel.add(n2); n2.addActionListener(this);
        n3=new JButton("3"); panel.add(n3); n3.addActionListener(this);
        minus=new JButton("-"); panel.add(minus); minus.addActionListener(this);

        dot=new JButton("."); panel.add(dot); dot.addActionListener(this);
        n0=new JButton("0"); panel.add(n0); n0.addActionListener(this);
        equal=new JButton("="); panel.add(equal); equal.addActionListener(this);
        plus=new JButton("+"); panel.add(plus); plus.addActionListener(this);

        add(panel, BorderLayout.CENTER);
    }

    public void actionPerformed(ActionEvent ae)
    {
        if(ae.getSource()==n1) assign("1");
        else if(ae.getSource()==n2) assign("2");
        else if(ae.getSource()==n3) assign("3");
        else if(ae.getSource()==n4) assign("4");
        else if(ae.getSource()==n5) assign("5");
        else if(ae.getSource()==n6) assign("6");
        else if(ae.getSource()==n7) assign("7");
        else if(ae.getSource()==n8) assign("8");
        else if(ae.getSource()==n9) assign("9");
        else if(ae.getSource()==n0) assign("0");

        else if(ae.getSource()==dot)
        {
            if(result.getText().indexOf(".") == -1)
```

```
            result.setText(result.getText() + ".");
        }

        else if(ae.getSource()==minus)
        {
            preRes = Double.parseDouble(result.getText());
            lastCommand = "-";
            result.setText("0");
        }

        else if(ae.getSource()==div)
        {
            preRes = Double.parseDouble(result.getText());
            lastCommand = "/";
            result.setText("0");
        }

        else if(ae.getSource()==mul)
        {
            preRes = Double.parseDouble(result.getText());
            lastCommand = "*";
            result.setText("0");
        }

        else if(ae.getSource()==plus)
        {
            preRes = Double.parseDouble(result.getText());
            lastCommand = "+";
            result.setText("0");
        }

        else if(ae.getSource()==equal)
        {
            secVal = Double.parseDouble(result.getText());

            if(lastCommand.equals("/")) res = preRes / secVal;
            else if(lastCommand.equals("*")) res = preRes * secVal;
            else if(lastCommand.equals("-")) res = preRes - secVal;
            else if(lastCommand.equals("+")) res = preRes + secVal;

            result.setText("" + res);
            lastCommand = "=";
        }
    }
}
```

2. a) JFrame Version — Simple Message Display

Program:-

```
import javax.swing.*;
import java.awt.*;

public class HelloJava extends JFrame
{
   public HelloJava()
   {
      setTitle("Hello Java");
      setSize(300, 200);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      add(new MessagePanel());

      setVisible(true);
   }

   public static void main(String[] args)
   {
      new HelloJava();
   }
}

class MessagePanel extends JPanel
{
   public void paint(Graphics g)
   {
      super.paint(g);
      g.drawString("Hello Java", 10, 100);
```

```
        }


    }
```

3**.b) JFrame Version — Factorial Calculator**

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Fact extends JFrame implements ActionListener
{
    JTextField t1, t2;
    JButton b0;

    public Fact()
    {
        setTitle("Factorial Calculator");
        setSize(350, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new GridLayout(3,2));

        add(new JLabel("Enter any Integer value"));
        add(t1 = new JTextField(20));

        add(new JLabel("Factorial value is:"));
        add(t2 = new JTextField(20));
        t2.setEditable(false);

        add(new JLabel(""));
        add(b0 = new JButton("Compute"));

        b0.addActionListener(this);

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e)
    {
        int n, f = 1;

        n = Integer.parseInt(t1.getText());

        for(int i = 1; i <= n; i++)
            f = f * i;

        t2.setText(String.valueOf(f));
    }

    public static void main(String[] args)
    {
        new Fact();
    }
}
```

3. Write a program that creates a user interface to perform integer divisions. The user enters two

numbers in the text fields, Num1 and Num2. The division of Num1 and Num2 is displayed in the Result field when the Divide button is clicked. If Num1 or Num2 were not an integer, the program would throw a NumberFormatException. If Num2 were Zero, the program would throw an Arithmetic Exception Display the exception in a message dialog box.

Program:-

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Add1 extends JFrame implements ActionListener
{
   JTextField num1, num2, res;
   JButton div;

   public Add1()
   {
      setTitle("Integer Division");
      setSize(350, 200);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      setLayout(new GridLayout(4,2));

      add(new JLabel("Number 1"));
      add(num1 = new JTextField(10));

      add(new JLabel("Number 2"));
      add(num2 = new JTextField(10));

      add(new JLabel("Result"));
      add(res = new JTextField(20));
      res.setEditable(false);

      add(new JLabel(""));
      add(div = new JButton("DIV"));
      div.addActionListener(this);
```

```
    setVisible(true);
}

public void actionPerformed(ActionEvent ae)
{
    try
    {
        int n1 = Integer.parseInt(num1.getText());
        int n2 = Integer.parseInt(num2.getText());

        int result = n1 / n2;

        res.setText(String.valueOf(result));
    }
    catch(NumberFormatException e)
    {
        JOptionPane.showMessageDialog(this,
                "NumberFormatException: Enter valid integers",
                "Error",
                JOptionPane.ERROR_MESSAGE);
    }
    catch(ArithmeticException e)
    {
        JOptionPane.showMessageDialog(this,
                "ArithmeticException: Cannot divide by zero",
                "Error",
                JOptionPane.ERROR_MESSAGE);
    }
}

public static void main(String[] args)
{
    new Add1();
}
```

}

5.) Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.

Program:-

```
class RandomGenThread implements Runnable
{
    double num; public void run()
    {
            try {
                    SquareThread sqt = new SquareThread();
                    Thread squareThread = new Thread(sqt);
                    CubeThread cbt = new CubeThread();
                    Threadcube Thread = new Thread(cbt);
                    squareThread.start();
                    cubeThread.start();
                    for(int i=0;i<10;i++)
                    {
                            System.out.println("t1-"+i);
                            if(i%2 == 0)
                            {
                            sqt.setNum(new Double(i));
                            }
                            else
                            {
                                    cbt.setNum(new Double(i));
                            }
                                    Thread.sleep(1000);
            }
            } catch (InterruptedException e)
                    {
                    e.printStackTrace();
                    }
        }

    }
    class SquareThread implements Runnable
    {
        Double num;
        public void run()
    {

            try {
```

```
                    int i=0;
                    do{
                        i++;
                        if(num != null&&num %2 ==0)
                         {
                        System.out.println("t2--->square of "+num+"="+(num*num));
                        num = null;
                        }
                    Thread.sleep(1000);
                    }while(i<=5);
                }
                     catch (Exception e)
                     {
                        e.printStackTrace();
                }
                }

    public Double getNum()
    {
            return num;
    }

    public void setNum(Double num)
    {
            this.num = num;
    }
}

class CubeThread implements Runnable
 {
    Double num;
    public void run()
    {
            try {
                    int i=0;
                    do{
                        i++;
                        if(num != null&&num%2 !=0)
                        {
                        System.out.println("t3-->Cube of "+num+"="+(num*num*num));
                        num=null;
                    }
                    Thread.sleep(1000);
                    }
                    while(i<=5);
```

```
                }
                        catch (Exception e)
                        {
                          e.printStackTrace();
                }
                }
        public Double getNum()
        {
                return num;
        }

        public void setNum(Double num)
        {
                this.num = num;
        }
    }
    public class MultiThreaded
     {
        public static void main(String[] args) throws InterruptedException
    {
                Thread randomThread = new Thread(new RandomGenThread());
                randomThread.start();
        }

    }
```

6) Write a java program that simulates a traffic light. The program lets the user select one of three lights: red, yellow, or green with radio buttons. On selecting a button, an appropriate message with "stop" or "ready" or "go" should appear above the buttons in a selected color. Initially there is no message shown.

Program:-

```
import javax.swing.*;
import java.awt.*;

public class TrafficSignal extends JFrame
{
   public TrafficSignal()
   {
      setTitle("Traffic Signal Simulation");
      setSize(350, 500);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      add(new SignalPanel());

      setVisible(true);
   }

   public static void main(String[] args)
   {
      new TrafficSignal();
   }
```

```
}

class SignalPanel extends JPanel implements Runnable
{
    Thread t;
    Font f, f1;
    int i = 0, a = 0, j = 0;

    public SignalPanel()
    {
        setBackground(Color.lightGray);
        f = new Font("TimesNewRoman", Font.ITALIC, 28);
        f1 = new Font("TimesNewRoman", Font.BOLD | Font.ITALIC, 28);

        t = new Thread(this);
        t.start();
    }

    public void run()
    {
        while(true)
        {
            for (i = 10; i >= 0; i--) // countdown
            {
                try { Thread.sleep(1000); }
                catch (Exception e) { }

                if (i <= 10 && i > 3)      // RED
                    a = 1;
                else if (i <= 3 && i > 0) // YELLOW
                    a = 2;
                else if (i == 0)          // GREEN
                {
                    for (j = 0; j < 10; j++)
                    {
                        a = 3;
                        try { Thread.sleep(1000); }
                        catch (Exception e) { }
                        repaint();
                    }
                }
                repaint();
            }
        }
    }

    public void paintComponent(Graphics g)
    {
        super.paintComponent(g);

        // Pole
        g.setColor(Color.black);
        g.fillRect(150, 150, 50, 150);
        g.fillRect(165, 300, 20, 155);
```

```
    // Signal outlines
    g.drawOval(150, 150, 50, 50); // Red
    g.drawOval(150, 200, 50, 50); // Yellow
    g.drawOval(150, 250, 50, 50); // Green

    // Countdown
    g.setColor(Color.red);
    g.setFont(f);
    g.drawString("" + i, 50, 50);

    // RED
    if (a == 1)
    {
      g.setColor(Color.red);
      g.fillOval(150, 150, 50, 50);
      g.drawString("STOP", 50, 150);
    }

    // YELLOW
    if (a == 2)
    {
      g.setColor(Color.yellow);
      g.fillOval(150, 200, 50, 50);
      g.drawString("READY", 50, 200);
    }

    // GREEN
    if (a == 3)
    {
      g.setColor(Color.blue);
      g.drawString("" + j, 150, 50);

      g.setColor(Color.green);
      g.fillOval(150, 250, 50, 50);
      g.drawString("GO", 50, 250);
    }
  }
```

7) Write a java program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contain only the method printArea( ) that prints the area of the given shape.

<u>Program:-</u>

```
abstract class Shape

{

 abstract void numberOfSides();

  }

class Trapezoid extends Shape
```

```
{
void numberOfSides()
 {
        System.out.println(" Trapezoidal has four sides");
}
  }
class Triangle extends Shape
{
 void numberOfSides()
{
 System.out.println("Triangle has three sides");
}
  }
 class Hexagon extends Shape
{
void numberOfSides()
{
System.out.println("Hexagon has six sides");
 }
 }
class ShapeDemo
```

```
{
 public static void main(String args[ ])
 {
Trapezoid t=new Trapezoid();
Triangle r=new Triangle();
Hexagon h=new Hexagon();
Shape s;
s=t;
s.numberOfSides();
s=r;
s.numberOfSides();
s=h;
s.numberOfSides();
}
}
```

8) Suppose that a table named Table.txt is stored in a text file. The first line in the file header and the remaining lines correspond to row in the table. The elements are separated by commas. Write a Java program to display the table using labels in grid layout.

Program:-

```
import java.awt.*;
import javax.swing.*;
import java.io.*;
import java.util.*;

public class Table1 extends JFrame
{
   Object data[][] = new Object[50][4];

   public Table1()
   {
      setTitle("Table Display from File");
      setSize(500, 300);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      Container con = getContentPane();
      con.setLayout(new BorderLayout());

      String[] colHeads = {"Name","Roll Number","Department","Percentage"};

      int i = 0;

      try
      {
         String fileName = JOptionPane.showInputDialog(
               "Enter the file name present in current directory");

         BufferedReader br = new BufferedReader(new FileReader(fileName));

         String line;
```

```
          while((line = br.readLine()) != null)
          {
             StringTokenizer st = new StringTokenizer(line, ",");

             int j = 0;
             while(st.hasMoreTokens())
             {
                data[i][j] = st.nextToken();
                j++;
             }
             i++;
          }
          br.close();
       }
       catch (Exception e)
       {
          JOptionPane.showMessageDialog(this,
                "Error: " + e.getMessage());
       }

       JTable table = new JTable(data, colHeads);

       JScrollPane scroll = new JScrollPane(table,
             JScrollPane.VERTICAL_SCROLLBAR_AS_NEEDED,
             JScrollPane.HORIZONTAL_SCROLLBAR_AS_NEEDED);

       con.add(scroll, BorderLayout.CENTER);

       setVisible(true);
    }

    public static void main(String[] args)
    {
       new Table1();
    }
}
```

10. Write a Java program that handles all mouse events and shows the event name at the center of the window when a mouse event is fired. (Use adapter classes).

Program:-

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class MouseDemo extends JFrame
{
   public MouseDemo()
   {
      setTitle("Mouse Event Demo");
      setSize(400, 300);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      add(new MousePanel());

      setVisible(true);
   }

   public static void main(String[] args)
   {
      new MouseDemo();
   }
}

class MousePanel extends JPanel
{
   int mx = 0, my = 0;
   String msg = "";

   public MousePanel()
   {
      // Mouse Listener using Adapter
      addMouseListener(new MouseAdapter()
      {
         public void mouseClicked(MouseEvent e)
         {
            msg = "Mouse Clicked";
            repaint();
         }

         public void mousePressed(MouseEvent e)
         {
            msg = "Mouse Pressed";
            repaint();
         }

         public void mouseReleased(MouseEvent e)
         {
            msg = "Mouse Released";
            repaint();
         }
```

```
            public void mouseEntered(MouseEvent e)
            {
                msg = "Mouse Entered";
                repaint();
            }

            public void mouseExited(MouseEvent e)
            {
                msg = "Mouse Exited";
                repaint();
            }
        });

        // Mouse Motion Listener using Adapter
        addMouseMotionListener(new MouseMotionAdapter()
        {
            public void mouseDragged(MouseEvent e)
            {
                mx = e.getX();
                my = e.getY();
                msg = "Mouse Dragged";
                repaint();
            }

            public void mouseMoved(MouseEvent e)
            {
                mx = e.getX();
                my = e.getY();
                msg = "Mouse Moved";
                repaint();
            }
        });
    }

    protected void paintComponent(Graphics g)
    {
        super.paintComponent(g);

        g.drawString("Handling Mouse Events", 120, 30);

        // Draw message at center
        FontMetrics fm = g.getFontMetrics();
        int x = (getWidth() - fm.stringWidth(msg)) / 2;
        int y = getHeight() / 2;

        g.drawString(msg, x, y);
    }
}
```

11).Write a java program that loads names and phone numbers from a text file where the data is organized as one line per record and each field in a record are separated by a tab (\t).it takes a name or phone number as input and prints the corresponding other value from the hash table(hint: use hash tables)

Program:-

```
import java.io.BufferedReader;

import java.io.File;

import java.io.FileNotFoundException;

import java.io.FileReader;

import java.io.IOException;

import java.util.Hashtable;

import java.util.Iterator;

import java.util.Set;

public class HashTab

 {

	public static void main(String[] args)

	{

		HashTab prog11 = new HashTab();

		Hashtable<String, String>hashData = prog11.readFromFile("HashTab.txt");

		System.out.println("File data into Hashtable:\n"+hashData);

		prog11.printTheData(hashData, "vbit");

		prog11.printTheData(hashData, "123");

		prog11.printTheData(hashData,  " ----");

	}

	private void printTheData(Hashtable<String, String>hashData, String input)

 {

		String output = null;

		if(hashData != null)
```

```
{
        Set<String> keys = hashData.keySet();
        if(keys.contains(input))
 {
                output = hashData.get(input);
        }
 else
 {
                Iterator<String> iterator = keys.iterator();
                while(iterator.hasNext()) {
                        String key = iterator.next();
                        String value = hashData.get(key);
                        if(value.equals(input))
                        {
                                output = key;
                                break;
                        }       }       }   }
System.out.println("Input given:"+input);
if(output != null)
 {
        System.out.println("Data found in HashTable:"+output);
        }
else {
        System.out.println("Data not found in HashTable");
}       }
```

```
private Hashtable<String, String>readFromFile(String fileName) {
    Hashtable<String, String> hashData = new Hashtable<String, String>();
    try {
        File f = new File("D:\\java\\"+fileName);
        BufferedReader br = new BufferedReader(new FileReader(f));
        String line = null;
        while((line = br.readLine()) != null) {
            String[] details = line.split("\t");
            hashData.put(details[0], details[1]);
        }
    } catch (FileNotFoundException e) {
        e.printStackTrace();
    } catch (IOException e) {
        e.printStackTrace();        }
    return hashData;      }  }
```

HashTab.txt

```
vbit    123
abc     345
edrf    567
```

1. Write a Java program that prints all real solutions to the quadratic equation $ax^2+bx+c = 0$. Read in a, b, c and use the quadratic formula. If the discriminate b^2-4ac is negative, display a message stating that there are no real solutions.

Program :-

```
import java.io.*;
class Quadratic
{
public static void main(String args[])throws IOException
{
double x1,x2,disc,a,b,c;
InputStreamReader obj=new InputStreamReader(System.in);
BufferedReader br=new BufferedReader(obj);
System.out.println("enter a,b,c values");
a=Double.parseDouble(br.readLine());
b=Double.parseDouble(br.readLine());
c=Double.parseDouble(br.readLine());
disc=(b*b)-(4*a*c);
if(disc==0)
{
System.out.println("roots are real and equal ");
x1=x2=-b/(2*a);

System.out.println("roots are "+x1+","+x2);
}
else if(disc>0)
{
 System.out.println("roots are real and unequal");
x1=(-b+Math.sqrt(disc))/(2*a);
x2=(-b+Math.sqrt(disc))/(2*a);
```

```
System.out.println("roots are "+x1+","+x2);
}
else
{
 System.out.println("roots are imaginary");
}   }   }
```

12) The Fibonacci sequence is defined by the following rule. The first 2 values in the sequence are 1, 1. Every subsequent value is the sum of the 2 values preceding it. Write a Java program that uses both recursive and non-recursive functions to print the n^th value of the Fibonacci sequence.

Program:-

```
/*Non Recursive Solution*/
    import java.util.Scanner;
    class Fib {
        public static void main(String args[ ]) {
                Scanner input=new Scanner(System.in);
                int i,a=1,b=1,c=0,t;
                System.out.println("Enter value of t:");
                t=input.nextInt();
                System.out.print(a);
                System.out.print(" "+b);
                for(i=0;i<t-2;i++) {
                        c=a+b;
                        a=b;
                        b=c;
                        System.out.print(" "+c);
                }
                System.out.println();
```

```
        System.out.print(t+"th value of the series is:  "+c);
    }
}
import java.io.*;

import java.lang.*;


    class Demo {
        int fib(int n) {
            if(n==1)
                return (1);

            else if(n==2)
                return (1);

            else
                return  (fib(n-1)+fib(n-2));
        }
    }

    class RecFibDemo {
    public static void main(String args[])throws IOException {
        InputStreamReader obj=new InputStreamReader(System.in);
        BufferedReader br=new BufferedReader(obj);
        System.out.println("enter last number");
        int n=Integer.parseInt(br.readLine());
        Demo ob=new Demo();
        System.out.println("fibonacci series is as follows");
        int res=0;
```

```
		for(int i=1;i<=n;i++) {
			res=ob.fib(i);
			System.out.println(" "+res); }
		System.out.println();
		System.out.println(n+"th value of the series is "+res);
		} }
```

13) **WAJP that prompts the user for an integer and then prints out all the prime numbers up to that Integer.**

Program:-

```
import java.util.*
class Test {
	void check(int num) {
		System.out.println ("Prime numbers up to "+num+" are:");
		for (int i=1;i<=num;i++)
			for (int j=2;j<i;j++) {
			  if(i%j==0)
			     break;
			  else  if((i%j!=0)&&(j==i-1))
			     System.out.print("  "+i);
			}
		}
}  //end of class Test
class Prime {
	public static void main(String args[ ]) {
		Test obj1=new Test();
		Scanner input=new Scanner(System.in);
		System.out.println("Enter the value of n:");
		int n=input.nextInt();
		obj1.check(n);
```

```
} }
```

14) WAJP that checks whether a given string is a palindrome or not. Ex: MADAM is a palindrome.

Program:-

```
import java.io.*;
class Palind {
    public static void main(String args[ ])throws IOException {
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
        System.out.println("Enter the string to check for palindrome:");
        String s1=br.readLine();
        StringBuffer sb=new StringBuffer();
        sb.append(s1);
        sb.reverse();
        String s2=sb.toString();
        if(s1.equals(s2))
                System.out.println("palindrome");
        else
                System.out.println("not  palindrome");
                }  }
```

15)WAJP for sorting a given list of names in ascending order.

Program:-

```
import java.io.*;
class Test {
    int len,i,j;
    String arr[ ];

    Test(int n) {
        len=n;
```

```
        arr=new String[n];
    }

    String[ ] getArray()throws IOException {
        BufferedReader br=new BufferedReader (new InputStreamReader(System.in));
        System.out.println ("Enter the strings U want to sort----- ");
        for (int i=0;i<len;i++)
            arr[i]=br.readLine();
        return arr;
    }

    String[ ] check()throws ArrayIndexOutOfBoundsException          {
        for (i=0;i<len-1;i++) {
            for(int j=i+1;j<len;j++) {
                if ((arr[i].compareTo(arr[j]))>0) {
                    String s1=arr[i];
                    arr[i]=arr[j];
                    arr[j]=s1;
                }
            }
        }
        return arr;
    }
    void display()throws ArrayIndexOutOfBoundsException {
        System.out.println ("Sorted list is---");
        for (i=0;i<len;i++)
            System.out.println(arr[i]);
        }
```

```
}     //end of the Test class

class Ascend {

    public static void main(String args[ ])throws IOException {

        Test obj1=new Test(4);

        obj1.getArray();

        obj1.check();

        obj1.display();

    }

}
```

WAJP to multiply two given matrices.

```
import java.util.*;

class Test {

        int r1,c1,r2,c2;

Test(int r1,int c1,int r2,int c2) {

                this.r1=r1;

                this.c1=c1;

                this.r2=r2;

                this.c2=c2;

        }

        int[ ][ ] getArray(int r,int c) {

                int arr[][]=new int[r][c];

                System.out.println("Enter the elements for "+r+"X"+c+" Matrix:");

                Scanner input=new Scanner(System.in);

                for(int i=0;i<r;i++)

                for(int j=0;j<c;j++)

                        arr[i][j]=input.nextInt();

                return arr;

        }

        int[ ][ ] findMul(int a[ ][ ],int b[ ][ ]) {
```

```
            int c[][]=new int[r1][c2];
            for (int i=0;i<r1;i++)
                for (int j=0;j<c2;j++) {
                    c[i][j]=0;
                    for (int k=0;k<r2;k++)
                        c[i][j]=c[i][j]+a[i][k]*b[k][j];
                }
            return c;
        }
        void  putArray(int res[ ][ ]) {
            System.out.println ("The resultant "+r1+"X"+c2+" Matrix is:");
            for (int i=0;i<r1;i++) {
               for (int j=0;j<c2;j++)
                    System.out.print(res[i][j]+"  ");
               System.out.println();
            } }  }   //end of Test class
class MatrixMul {
        public static void main(String args[ ])throws IOException {
            Test obj1=new Test(2,3,3,2);
            Test obj2=new Test(2,3,3,2);
             int x[ ][ ],y[ ][ ],z[ ][ ];
             System.out.println("MATRIX-1:");
            x=obj1.getArray(2,3);         //to get the matrix from user
            System.out.println("MATRIX-2:");
            y=obj2.getArray(3,2);
            z=obj1.findMul(x,y);     //to perform the multiplication
            obj1.putArray(z);          // to display the resultant matrix
        } }
```

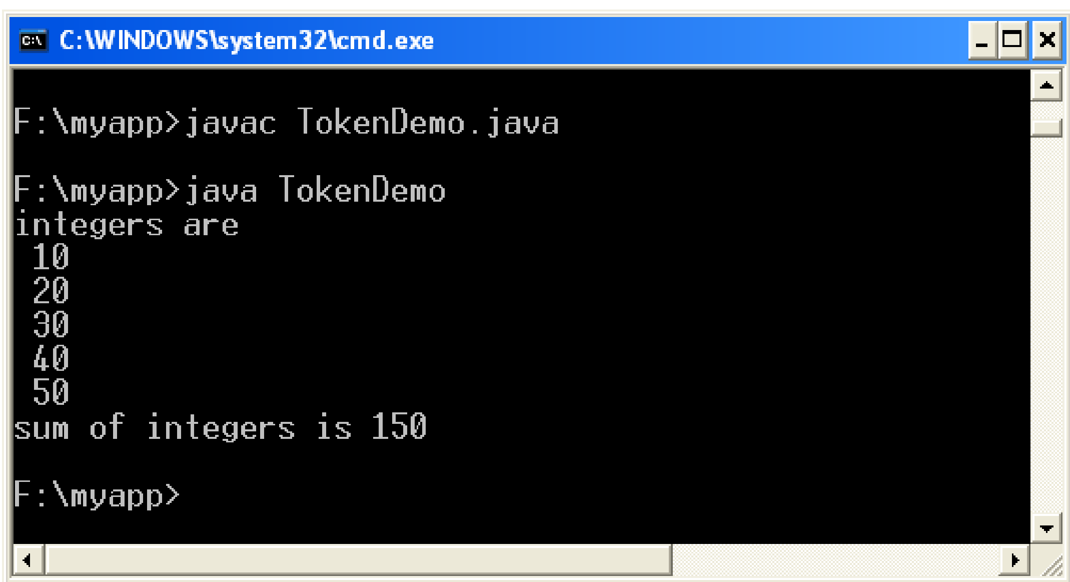
16) WAJP that reads a line of integers and then displays each integer and the sum of all integers. (use StringTokenizer class)

Program :-

```
// Using StringTokenizer class

import java.lang.*;

import java.util.*;

class tokendemo {

    public static void main(String args[ ]) {

        String s="10,20,30,40,50";

        int sum=0;

        StringTokenizer a=new StringTokenizer(s,",",false);

        System.out.println("integers are ");

        while(a.hasMoreTokens()) {

                int b=Integer.parseInt(a.nextToken());

                sum=sum+b;

                System.out.println("  "+b);

        }

        System.out.println("sum of integers is "+sum);

    }

    }

// Alternate solution using command line arguments

class Arguments {

        public static void main(String args[ ]) {

                 int sum=0;

                  int n=args.length;

                 System.out.println("length is "+n);


                int arr[]=new int[n];
```

```
for(int i=0;i<n;i++)
arr[i]=Integer.parseInt(args[i]);
System.out.println("The enterd values are:");
for(int i=0;i<n;i++)
System.out.println(arr[i]);
System.out.println("sum of enterd integers is:");
for(int i=0;i<n;i++)
sum=sum+arr[i];
System.out.println(sum);
}
}
```

Input & Output :-



17) WAJP that displays the number of characters, lines and words in a text file.

Program :-

```
import java.io.*;
public class FileStat {
    public static void main(String args[ ])throws IOException {
        long nl=0,nw=0,nc=0;
        String line;
        BufferedReader br=new BufferedReader(new FileReader(args[0]));
        while ((line=br.readLine())!=null) {
            nl++;
            nc=nc+line.length();
            int i=0;
            boolean pspace=true;
            while (i<line.length()) {
                char c=line.charAt(i++);
                boolean cspace=Character.isWhitespace(c);
                if (pspace&&!cspace)
                    nw++;
                pspace=cspace;
            }
        }
        System.out.println("Number of Characters"+nc);
        System.out.println("Number of Characters"+nw);
        System.out.println("Number of Characters"+nl);
}}

// Alternate solution using StringTokenizer
import java.io.*;
```

```
import java.util.*;

public class FileStat {

    public static void main(String args[ ])throws IOException {

        long nl=0,nw=0,nc=0;

        String line;

        BufferedReader br=new BufferedReader(new FileReader(args[0]));

        while ((line=br.readLine())!=null) {

            nl++;

            nc=nc+line.length();

            StringTokenizer st = new StringTokenizer(line);
            nw += st.countTokens();
        }
        System.out.println("Number of Characters"+nc);
        System.out.println("Number of Characters"+nw);

        System.out.println("Number  of  Characters"+nl);

 }}
```

<u>Input & Output :-</u>